INCH-POUND
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(See 6.12)

MILITARY SPECIFICATION

DISTILLATION UNIT, WATER, THERMOCOMPRESSION

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electrically operated distillation units of the thermocompression type for use on Naval ships.

1.2 Classification.

1.2.1 Types. Distillation units shall be of the following types as specified (see 6,2):

Type I - Flooded tube evaporator design.
Type II - Rising film evaporator design.

1.2.2 Classes. Distillation units shall be of the following classes as specified (see 6.2):

Class A - Basically composition 70-30 copper-nickel alloy for submarine application (see 3.2.1) .
Class B - Basically composition 90-10 copper-nickel alloy for surface ship application (see 3.2.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4620

AMSC N/A

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

TT-P-28	- Paint, Aluminum, Heat Resisting (1200°F).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-C-104	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	Preservation, Methods of.
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-C-132	- Crate, Wood, Open; Maximum Capacity 2,500 Pounds.
MIL-S-901	- Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-A-907	- Antiseize Thread Compound, High Temperature.
MIL-C-915	- Cable and Cord, Electrical, for Shipboard Use, General Specification for.
MIL-E-917	- Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-F-1183	- Fittings, Pipe, Last Bronze, Silver-Brazing, General Specification for.
MIL-F-1183/10	- Fittings, Pipe, Cast Bronze, Silver-Brazing; Union, Pipe,
MIL-S-1222	- Studs, Bolts, Hex Cap Screws, Socket Head Cap Screws and Nuts,
MIL-E-2036	- Enclosures for Electric and Electronic Equipment.
MIL-V-2042	- Valves Reducing, Water Service, for Naval Shipboard Use.
MIL-C-2212	- Controllers, Electric Motor A.C. or D C., and Associated Switching Devices.
MIL-G-2860	- Glasses, Sight-Flow Clear, Borosilicate.
MIL-C-3774	- Crates, Wood; Open 12,000- and 16,000-Pound Capacity.
MIL-P-5425	- Plastic Sheet, Acrylic, Heat Resistant,
MIL-C-11029	- Critic Acid, Technical.
MIL-T-15005	- Tubes, 70-30 and 90-10 Copper Nickel Alloy, Condenser and Heat Exchanger.
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment.

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MILITARY (Continued)

MIL-P-15024/5 - Plates, Identification.
MIL-S-15103 Salinity Indicating Equipment.
MIL-E-15465 - Ejector Assemblies, Air.
MIL-C-15726 Copper-Nickel Alloy, Rod, Flat Produces (Flat Wire, Strip, Sheet, Bar, and Plate) and Forgings.
MIL-C-15730 - Coolers, Fluid, Naval Shipboard; Lubricating Oil. Hydraulic Oil, and Fresh Water.
MIL-E-15809 - Expander, Tube, Condenser, and Heat Exchangers.
MIL-T-16420 - Tube, Copper-Nickel Alloy, Seamless and welded (Copper Alloy Numbers 715 and 706).
MIL-V-16556 - Valve, Solenoid, Three-Way Bypass (Naval Shipboard Use).
MIL-M-17060 - Motors, 60-Hertz, Alternating Current, Integral-Horsepower, Shipboard Use.
MIL-I-17244 - Indicators, Temperature, Direct-Reading, Bimetallic, (3 and 5 Inch Dial).
MIL-M-17413 - Motors, Direct Current, Integral H.P. , Naval Shipboard.
MIL-E-17639 - Pumps, Centrifugal, Miscellaneous Service, Naval Shipboard Use.
MIL-P-17840 - Pumps, Centrifugal, Close-coupled, Navy Standard (for Surface Ship Application).
MIL-S-17849 - Strainers, Sediment, Pipeline, Duplex (With and Without Magnet).
MIL-B-17931 - Bearings, Ball, Annular, for Quiet Operation.
MIL-P-18472 - Pumps, Centrifugal, Condensate, Feed Booster and Distilling Plant.
MIL-G-18997 - Gauge, Pressure, Dial Indicating.
MIL-T-19646 - Thermometers, Remote Reading, Self-Indicating Dial, Gas Actuated.
MIL-I-20037 - Indicators, Sight , Liquid Level Direct/Indirect Reading, Tubular Glass/Plastic.
MIL-F-20042 - Flanges, Pipe and Bulkhead, Bronze (Silver Brazing).
MIL-R-21252 - Rubber Sheet, Solid, synthetic, shipboard Water Evaporator Gasketing.
MIL-G-22050 Gasket and Packing Material, Rubber, For Use With Polar Fluids, Steam, and Air at Moderately High Temperatures .
MIL-S-22473 - Sealing, Locking, and Retaining Compounds; (Single-Component) .
MIL-H-22577 - Heating Elements, Electrical: Cartridge, Strip and Tubular Type.
MIL-L-24131 - Lubricant, Colloidal Graphite in Isopropanol.

STANDARDS

FEDERAL

FED-STD-H28 Screw Thread Standards for Federal Services.

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MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-167-1	- Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-278	Welding and Casting Standard.
MIL-STD-438	- Schedule of Piping, Valves, Fittings, and Associated Piping Components for Submarine Service .
MIL-STD-740-1	- Airborne Sound Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-740-2	- Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-769	Thermal Insulation Requirements for Machinery and Piping,
MIL-STD-777	- Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships.
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production.
MIL-STD-1399 Section 300	- Interface Standard for Shipboard Systems Electric Power, Alternating Current. (Metric)
DOD-STD-1399 Section 301	Interface Standard for Shipboard Systems Ship Motion and Attitude, (Metric)
MIL-HDBK-267	· Guide for Selection of Lubricants and Hydraulic Fluids for Use in Shipboard Equipment.

(Unless otherwise indicated, copies of federal and military specifications standards , and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2,1,2 Other Government drawing. The following Government drawing forms a part of this document to the extent specified herein. Unless otherwise specified, the issue is that cited in the solicitation.

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

B-214	- Root Connections for Attaching Piping.

(Application for copies should be addressed to the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B1.12 - Class 5 Interference Fit Thread.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 395 - Standard Specification for Ferritic Ductile Iron Pressure - Retaining Castings for Use at Elevator Temperatures.

B 265 - Standard Specification for Titanium and Titanium Alloy Strip, Sheet, and Plate. (DoD adopted)

B 338 - Standard Specification for Seamless and Welded Titanium and Titanium Alloy Tubes for Condensers and Heat Exchangers, (DoD adopted)

B 552 - Standard Specification for Seamless and Welded Copper/Nickel Tubes for Water Desalting Plants.

D 512 - Standard Test Methods for Chloride Ion in Water. (DoD adopted)

D 3951 - Standard Practice for Commercial Packaging (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.2 Materials. The materials shown in table I shall be used as the basis for design, They shall be used in the construction of the distillation units except that the manufacturer shall have the option of substituting commercial material having equal or better physical and chemical properties than the materials specified under Federal or Military specifications. Proposed alternate materials shall be subject to approval by the design review activity

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TABLE 1. Materials for evaporators and heat exchangers.

Application	Material	Applicable document
Evaporators:		
Shell and baffles	Copper-nickel alloy,	MIL-C-15726
Tubes and tubesheets 1/	Copper-nickel alloy,	MIL-T-15005
	Titanium, grade 2	MIL-C-15726
		ASTM B 338
Tubing	Copper-nickel alloy,	MIL-T-16420
Tubing, fluted	Copper-nickel alloy,	ASTM B 552
Gaskets	Synthetic rubber	MIL-R-21252
		MIL-G-22050
Piping	Cu-Ni	MIL-STD-438
		MIL-STD-777
Shell and tube heat exchangers :		
Shell, baffles and tube sheets	Copper-nickel alloy	MIL-C-15726
		MIL-T-16420
Tubes	Copper-nickel alloy	MIL-T-15005
Gaskets	Synthetic rubber	MIL-R-21252
		MIL-G-22050
Plate type heat exchangers :		
Plates	Titanium, grade 1 or 2	ASTM B 265
Frame	Carbon steel	Commercial
Gaskets	EPDM	Commercial
Bolts, studs, and nuts for all components:	Nickel-copper alloy,	MIL-S-1222
Tubing, air lines	Copper-nickel	MIL-T-16420
Deaerator	Glass reinforced plastic (Class B only)	Commercial
	Copper-nickel alloy	MIL-C-15726
Deaerator After condenser	Copper-nickel alloy (shell and tubes)	MIL-E-15465
Vent condenser	Copper-nickel	MIL-C-15726

1/ Tubes and tubesheets shall be fabricated from the same material to preclude galuonic corrosion.

3.2.1 Class A distiller materials. Materials shall be in accordance with table X. Copper-nickel alloy shall be composition 70-30.

3.2.2 Class B distiller materials. Materials shall be in accordance with table I, Copper-nickel alloy shall be composition 90-10.

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3.2.3 Mercury exclusion. Distillation plants shall not contain mercury in any form. Mercury-bearing instruments or equipment which might cause contamination shall not be used in the manufacture, fabrication, assembly, or testing of any material. In the event of any accident involving mercury contamination Of the q aterial being furnished or suspicion of such contamination, NAVSEA or the activity concerned shall be notified immediately.

3.2.4 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term “recovered materials” means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2.5 Cadmium exclusion. Cadmium plated parts shall not be used,

3.2.6 Lubricants and compounds. Distilling plants shall be designed to use only those lubricants or compounds listed in MIL-HDBK-267, for shipboard maintenance (see 6.9).

3.2.7 Safety. Equipment shall ensure safety to operating and maintenance personnel. When the equipment is properly installed and the enclosure is grounded, there shall be no accessible way for operating personnel to receive an electric shock even though an internal fault between any two circuits, between any circuit and a structural member, or between any circuit and ground may exist. The design shall hold to a practical minimum the possibility of maintenance personnel being exposed to electric shock while servicing, adjusting or checking out the equipment. External moving parts which are a potential hazard to personnel shall be avoided. When their use is unavoidable, positive protection in the form of a guard shall be provided. Sharp corners and projections which may cause injury in a rough sea on to which clothing may catch shall be avoided. Equipment shall be designed to revert to its least hazardous condition or mode of operation upon failure of a circuit or part. Special performance capabilities (continuation of operation, automatic shut-down or any other specific function) in the event of an electrical part failure shall be as required by the equipment specification (see 3.3.8). The design shall be such that should failure occur, any resulting damage will be confined to the smallest equipment subdivision (minimum replaceable part or subassembly) within which failure occurred.

3.2.8 Galvanic Protection, Direct contact of dissimilar materials shall be avoided, If direct contact cannot be avoided, galvanic protection shall be added in the form of sacrificial anodes.

3.3 Design. The distillation unit shall be of the vapor compression type (thermocompression) , wherein electrical power is the only source of power for the operation. The compressor shall be electric-motor driven. Heat shall be provided by electric immersion heaters. The units shall be furnished either “knocked-down” in several component parts, with all external piping and wiring to

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be installed by the shipyard, or as a self-contained “package type” unit ready for operation after installation as a unit by the shipyard, as specified (see 6.2). Unless otherwise specified, type II distillers shall be “package” type only. The base of package type units shall be suitable for resilient mounting to the ship’s structure using resilient mounts as specified (see 6.2). The units shall consist of the following:

(a) One evaporator.
(b) One or more heat exchangers, the primary heat exchanger for heating the feed and cooling brine and distillate shall be of the plate type.
(c) Equipment to keep the evaporator free from noncondensible gases.
(d) One centrifugal vapor compressor, with pulley and belts, if required, and motor and controller.
(e) Distillate pump in accordance with MIL-P-18472.
(f) Brine pumps in accordance with MIL-P-18472.
(g) When specified (see 6.2), feed pump to supply designated feedwater pressure in accordance with MIL-P-17840 or MIL-P-17639 as applicable.
(h) Salinity indicating equipment in accordance with MIL-S-15103.
(i) Solenoid operated, three way bypass valve in accordance with MIL-V-16556.
(j) A distillate totalizing meter, feed and distillate flowmeters or rotameters as approved by the contracting activity or Government agency, thermometers, and duplex feedwater strainer in accordance with MIL-s-17849.
(k) One set of accessories to complete the unit including pressure gauges in accordance with MIL-G-18997, ammeters (one for the electric heater circuit and one for the compressor motor circuit) , pilot light to indicate when heaters are operating, regulating flow control and relief valves, and electric immersion heaters complete with automatic and manual control switches.

3.3.1 Electric immersion heaters. Electric immersion heaters may be installed in a copper-nickel (see 3.2 and 3.2.2 for composition) tank separate from the evaporator. When installed in the evaporator, electric immersion heaters for stabilization of operation shall be installed in the distillate side of the evaporator in such manner that they will be immersed in distillate under all conditions of inclination of the ship (see 3.3.9). If no positive means are used to keep the heaters immersed in water for all ship motion and attitude, then a low water level shutdown device shall be provided to protect the heaters.

3,3.1.1 Additional requirements for electric immersion heaters are as follows:

(a) Additional electric immersion heaters, as required for satisfactory operation with feedwater between 28 and 85°F, shall be provided for installation in the feed or distillate line ahead of the heat exchangers. The heaters shall be provided with electric switches (see 3,15,1.2) thermostatically controlled, to turn the heaters off when the temperature of the feedwater is such that additional heating over that supplied by the stabilization

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heaters is not required. Heaters shall consist of tubular type electric heating elements in accordance with MIL-H-22577, secured to cast bronze flanges by brazing or compression fittings. The heating element sheath material shall be seamless and fabricated from Inconel 625 or Incology 800. Watt density shall not exceed 70 watts per square inch. The heater shall provide for minimum service life of 10,000 hours and 50,000 cycles. The terminal box shall be class 1, drip proof, in accordance with MIL-E-2036. The ends of the element shall be hermetically sealed by ceramic to metal seals. Each heating unit shall be provided with an identification plate giving connection data. The heaters shall be provided with temperature controllers to cycle the heater as needed.

3,3.1.2 Dielectric voltage. Heating units shall withstand, for a period of 1 minute, a dielectric test voltage of twice the rated voltage plus 1000 volts between the heating elements and ground, applied continuously for a period of not less than 60 seconds. The frequency shall be not less than 60 hertz (Hz) and the voltage wave shall approximate a sine wave.

3.3.1.3 Insulation resistance. The insulation resistance of each heating unit , when at normal operating temperatures shall be not less than 2.0 megohms. The insulation resistance at 80 ± 5 degrees Fahrenheit(°F) shall be not less than 4,0 megohms.

3.3,2 Flow control valves. The regulating flow control valve as specified in 3.3(k) shall be of the type which regulates the flow rate of feedwater to that required by the evaporator regardless of the initial pressure of the feedwater. Flow control valves shall be in accordance with MIL-V-2042.

3.3.3 Rotameters. Rotameters shall be of the type in which the float is guided by flutes in the tube. Meters shall be provided with metal casings with safety glass windows. Float tube packing gland followers, bolts, and nuts shall be nonferrous. The inlet and outlet shall be flanged.

3.3.4 Flow meters. Flow meters shall be provided for each distillation unit. Meters for flow of feed or brine shall not be readily affected by scale formation. In particular, meters equipped with a guide rod on which the float rides , or with an indicating rod mounted on the float and extending through a guide or packing gland are not acceptable. If method of feed is such that the ratio of feed and distillate must be controlled, feed and distillate flow meters shall be provided; otherwise, only a feed flow meter will be required.

3.3.5 Totalizing meters. A distillate totalizing meter shall be provided with each unit. The meter shall conform to MIL-F-1183 and MIL-F-1183/10 with magnetically driven totalizing register and with union connections (with retaining ring and O-ring seal).

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3,3.6 Thermometers. Provisions shall be made for installation of thermometers , as required. Unless otherwise specified (see 6.2), thermometers shall be furnished by the manufacturer. Thermometers shall be in accordance with MIL-I-17244. When location of thermometers would make it difficult to read, distant reading thermometers shall with design activity approvals be substituted. Distant reading thermometers shall be in accordance with MIL-T-19646.

3.3.7 Pressure damage. Class A distillation units shall not be damaged when subjected to ambient pressures of 30 pounds per square inch (lb/in*) absolute maximum or 10 lb/in^2 absolute minimum. The parts of the distilling units shall pass a hydrostatic pressure test *as* specified in 4.6.6.

3.3.8 Automatic operation. Distillation units shall provide for automatic operation to require minimum attention. Automatic operation is defined as the capability to start, stop, and place the distiller in operation from an operator's panel. Class A unit shall operate satisfactorily when the ambient pressure varies from normal atmospheric to plus or minus 6 inches of mercury. Distillation units shall be designed to shut down automatically when any of the following parameters occur.

(a) Compressor high oil temperature.
(b) Compressor low oil pressure.
(c) Evaporator high water level.
(d) Immersion heater (boiler section) low water level.

3.3.9 Ship attitude and motion. Unless otherwise specified (see 6.2), distillation units shall meet performance requirements of this specification when the units are operated under ship motion and attitude conditions specified in DOD-STD-1399, section 301, For submarine units, surfaced conditions shall also apply for the snorkel condition.

3.3.10 Deaeration. A suitable method to deaerate the evaporator shall be provided. The method and associated equipment designs shall be approved by the contracting activity.

3.4 Capacity and distillate Quality.

3.4.1 Capacity. Rated distillate capacity shall be as specified (see 6.2). Rated capacity is defined as the average rate of distillate produced over a period of 250 hours of operation when evaporators are supplied with seawater of not less than 35,000 parts per million (p/m) total dissolved solids (TDS) at feed temperatures from 28 to 85°F. Class A distillers shall also meet the capacity requirements with fluctuations of plus or minus 1500 ppm TDS under normal operating conditions of vertical travel and with a maximum transient temperature change of plus or minus 20°F in 2 minutes under normal operating conditions of vertical travel. This capacity may be achieved using either continuous or intermittent injection of feedwater treatment chemicals to control scaling as specified (see 3.4.2 and 6.2). Securing the distiller for cleaning is not permitted during the 250 hour period. Rated capacity shall be produced under the following conditions:

(a) Automatic operation in accordance with 3.3.8.
(b) Ship motion in accordance with 3.3.9.

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(c) Compressed vapor saturation temperature not to exceed 250°F.
(d) Temperature of distillate from the unit not to exceed 125°F unless otherwise specified (see 6.2).
(e) Concentration ratio not to exceed 1.8. Concentration ratio is defined as (feed flow rate) divided by (feed flow rate minus distillate rate).
(f) Quality of distillate in accordance with 3.4.3.

3,4,2 Scale control. The distiller design shall include provisions for on-line injection of feedwater chemicals to control scale formation and for off-line injection and circulation of NAVSEA approved acid solutions to remove built up scale which cannot be controlled by on-line methods. On-line is defined as not interrupting the production of distillate, that is, the distiller continues to make rated capacity and quality as defined in 3.4: Off-line is defined as corrective maintenance performed when the distiller is not being operated to produce distillate,

3.4.2.1 On-line injection of feed treatment chemicals to control scale formation. The On-line chemical feed treatment process shall not interfere with or reduce the rated capacity of the distiller, For class A units, the process may use either the continuous injection feed treatment system (see 3.4.2.1.2) or the Periodic acid shot method (see 3.4.2.1.1). Class B, type II units shall use the continuous injection system.

3,4.2.1.1 Periodic acid shot (PAS) feed treatment system. The PAS system provides for injection of an acid solution into the distilling plant seawater circuit at approximately 6 to 8 hour intervals. Each acid shot remove scale built up since the last injection. The acid solution shall be sodium bisulfate (see 6.10).

3,4.1.2 Continuous injection feed treatment system. The continuous injection feed treatment system provides continuous injection of NAVSEA approved feed treatment chemicals to retard scale formation. This system shall also provide for on-line injection of citric acid in accordance MIL-C-11029 to remove the sludge type scale formed by continuous injection of approved feed treatment chemicals . Continuous injection of acid solutions to control scale is prohibited (see 6.10). The use of sulfamic acid for on-line cleaning is prohibited whenever the distiller process temperatures exceed 150°F.

3.4.2.2 Off-line acid cleaning. In order to remove the scale that forms despite use of on-line treatment methods, the distiller design shall include provisions to inject and circulate acid solutions. The acid solutions shall be sulfamic acid with corrosion inhibiters (see 6.10). The acid solution shall be circulated through the evaporator, brine pump, feed heat exchangers and all other components and piping which are susceptible to scale formation. A method shall also be provided to heat the acid solution to a temperature of 120°F to increase acid reaction with the scale, Boiler section heaters may be used for this purpose. Since this is an off-line maintenance requirement, cross connecting piping and acid storage tanks and so forth, necessary to accomplish this function shall not be included with the distiller. Rather, adequate provision shall be made to provide connections, fittings, and so forth so that a maintenance activity may easily set Up the distiller for off-line cleaning.

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symmetrically with associated controls

3.4.2.3 Chemical tanks for continuous injection feedwater treatment chemicals, Tanks used for continuous injection feedwater treatment chemicals and acid shall be of sufficient size and access opening to allow mixing of chemicals inside the tank. Size of the access opening shall be approved by the design approval activity. A means shall be provided to determine the liquid level in the tank. This device shall be graduated in 1/2 gallon increments, Tanks and measuring devices shall meet section 3 requirements of this specification. The lid cover with quick release/latching device shall be watertight. The tanks shall be provided with a bottom drain valve. The tanks shall be made of the specified copper-nickel alloy. Tanks shall have a standpipe in the center of the tank and at a height to preclude sucking air during ship motion.

3.4.3 Distillate quality.

(a) Submarine or nuclear propelled surface ships. Salinity content of distillate produced shall not exceed 0,0325 equivalents per million (epm), 1.15 (p/m).
(b) Non-nuclear surface ships. Salinity content of distillate produced shall not exceed 0.065 epm, 2.32 p\m chlorides or 0.250 grams per gallon of sea salts.

3.4.4 Starting operation. The unit shall produce distillate meeting requirements as specified in 3.4.3 and 3.4.1 within 4 hours of start up for type I distillers and within 2 hours for type II distillers over a range of seawater temperatures of 28 to 85°F.

3.4.5 Clean tube operation.

(a) With clean heat transfer surfaces, class A units shall produce at least 110 percent of rated capacity and class B units shall produce 120 percent of rated capacity.
(b) Clean tube capacity in units using the continuous injection feed treatment system shall be achievable by on-line citric acid cleaning, The total time to start and complete an on-line acid cleaning cycle shall not exceed 2 hours.

3,4.6 Plugged tube operation. The unit shall produce rated capacity of distillate when 10 percent of the evaporator tubes are plugged,

3,5 Power. Distilling plant power requirements shall be tested as specified in 4.6.1.

3.6 Shock resistance, sound and vibration.

3.6.1 Shock. Unless otherwise specified (see 6,2), the assembled unit shall conform to grade A shock requirements in accordance with MIL-S-901. Shock tests of complete distillation units or of separately mounted components thereof weighing less than 15,000 pounds are required. Shock test extension will be considered for units if they are of similar design, construction, weight and materials to units which have successfully passed shock test in accordance with MIL-S-901. Shock test extension requests shall be prepared in accordance with MIL-S-901.

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3.6.1.1 Under the test procedure for medium weight equipment, the first blow in each group shall be applied using the standard horizontal mounting adaptor; the second blow in each group shall be applied with the unit mounted on adaptors holding it at a 30 degree angle from the horizontal in the direction of its least transverse strength.

3.6.1.2 The distillation unit, after sustaining mechanical shock, shall be capable of normal operation (see 3.3 and 3,4) and no parts thereof shall create a missile hazard to personnel or contiguous apparatus or otherwise constitute a personnel hazard (such as rupture of parts containing steam or hot water, with consequent danger of scalding personnel) at the time of the shock.

3.6.1.3 Support. Nozzle flanges or piping shall not constitute the means of support of components or accessories.

3.6.1.4 Bolts. Bolts designed to be stressed in shear shall be installed in holes not greater than the following sizes:

Nominal bolt diameter (inch)	Maximum diameter of hole (inch)
3/4 and smaller	Nominal bolt diameter plus 1/32
Larger than 3/4	Nominal bolt diameter plus 1/16

3.6.1.5 Units that are rigidly supported shall not be attached to two structures which can deflect relative to each other under shock loadings.

3.6.1.6 Shock mounts shall not be used without prior approval of the Command or agency concerned.

3.6.2 Noise and vibration. Unless otherwise specified (see 6.2), noise and vibration requirements shall be as specified herein.

3.6.2.1 Noise. All units shall meet the airborne noise requirements of grade D of MIL-STD-740-1. All compressors shall meet the MIL-STD-740-2 structureborne noise requirements for type I equipment and the pumps shall meet MIL-STD-740-2 requirement for type II equipment.

3.6 2.2 Vibration. The unit and all components shall be in accordance with type 1 and II vibration of MIL-sTD-167-1. Type IX requirements are through first order vibrations only.

3.6 2.3 When specified (see 6.2), ball bearings of rotating components of distillation units shall be in accordance with MIL-B-17931.

3.7 Evaporator.

3.7.1 Evaporator desire. The evaporator shall be composed of a shell, tube sheets, tubes and the necessary baffles and demisters to ensure delivery of distillate having the specified purity. The minimum shell thickness shall be 0.125 inch. Sight windows assemblies and level indicators as specified in 3.7,3 and 3.7,4 shall be provided as necessary to ensure water levels can be observed during operation and chemical cleaning.

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3.7.2 Boiler section. Distillate boiler section may be provided below the salt water feed and recirculation section or may be a separate component. It shall be provided with connections for insertion of the electric heating elements , distillate supply from the evaporator distillate collection space, vapor offtake connection to the evaporator section, a filling connection for start-up, and as necessary for operation, sight window assemblies and level indicators as specified in 3.7.3 and 3.7.4.

3.7,3 Sight window assemblies. Sight window assemblies using glass in accordance with MIL-G-2860 shall be composed of a metal flanged frame to be bolted to a boss or flange on the evaporator shell. The frame shall contain the sight window. The window shall be secured with its gaskets by means of a ring screwed into the frame to obtain even pressure on the window so that the pressure on the window is not changed when the sight window frame assembly is removed from the evaporator shell. Jack screws shall be provided in the flange of the frame.

3.7.3.1 Sight window assemblies using heat resistant, plastic acrylic sheet in accordance with MIL-P-5425, in lieu of glass, shall be composed of a single flange bolted to a boss or flange on the evaporator shell and confining the plastic sheet between resilient gaskets. This shall not be used in applications above 185°F. Jack screws shall be provided in the flange.

3.7.4 Level indicators. Evaporators shall have indicators of the indirect type in accordance with MIL-I-20037 to indicate water levels in the evaporator and boiler sections.

3.8 Feed heat exchangers. Unless otherwise specified (see 6.2), the feed heat exchangers shall be of the plate and frame type for class B units. For class A units, the feed heat exchanger shall be either shell and tube or plate and frame. It shall be a three stream design with the incoming feedwater heated by the exiting brine and distilled water. Plate and flame heat exchangers shall consist of corrugated gasketed plates compressed between a fixed end plate and a movable compression plate. Shell and tube units shall conform to MIL-C-1573G. Materials shall be specified in table I. Plate and frame heat exchangers shall be designed such that two men can disassemble, service, and assemble it in place.

3.9 Tubes and tube sheets for evaporator. When tubes are expanded into tube sheets or headers, the expansion shall be accomplished by means of an automatic tube expander control, using a tube expander in accordance with MIL-E-15809. The expander shall be adjusted so that the expansion is stopped not closer than 1/8 inch to the inner face of the tube sheet. Holes for copper-nickel tubes in the tube sheets or headers shall be drilled 0.002 inch larger than the nominal outside diameter of the tubes with plus 0.005 inch tolerance. Minimum ligament between holes for tubes shall be 3/16 inch. Minimum tube sheet thickness shall be 3/4 inch. Tube hole drilling requirements for titanium tubes shall be approved by the drawing review activity.

3.10 Compressor. Unless otherwise specified, the compressor shall be of the single stage centrifugal type. Materials shall be bronze, copper-nickel alloy, titanium, or other material suitable for the purpose intended. Compressor design and materials shall be approved by the contracting activity or Government

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agency. Compressor bearing and seals shall be designed for a minimum of 20,000 hours of operation. New compressor design shall undergo 1,000 hours of performance testing using steam at operating conditions.

3.10.1 Motors and controllers. Motors and controllers shall be furnished in accordance with MIL-M-17060 or MIL-M-17413 and MIL-C-2212, respectively. Alternating current (at) motors shall be in accordance with MIL-M-17060.

3.10.2 Motor insulation and enclosures. Motors shall be totally enclosed or totally enclosed fan cooled with sealed insulation in accordance MIL-M-17060.

3.10.3 Motor structureborne noise. Unless otherwise specified (see 6.2), structureborne noise levels shall be in accordance with MIL-STD-740-2 for type 111 equipment.

3.10.4 Motors air borne noise. Motor airborne noise shall not exceed the levels specified in table 11.

TABLE II. Airborne noise limits (sound pressure levels in decibels (dB) referenced to 20 micropascals (uPa).

Hp	Enclosure	Synchronous speed r/rein	Octaveband center frequency (Hz)								
			31.5	63	125	250	500	lK	2K	4K	8K
1 through 5	All	All	66	63	60	57	54	51	48	45	42
7.5 through 25	All	1800 and less	66	63	60	57	54	51	48	45	42
7.5 through 25	All	3600	66	63	60	66	65	64	63	57	54
30 through 150	All	1800 and less	66	66	66	69	69	63	63	56	53
30 through 150	All	3600	66	66	66	73	73	67	67	60	57
Above 150	All	All	66	66	66	73	73	67	67	60	57

3.11 Weight. The unit shall be designed to minimize operating and dry weights.

3.12 Access for installation. maintenance or replacement. The distilling unit shall be designed so that all parts requiring maintenance, cleaning, adjustment , or observation are readily accessible. The design shall incorporate features for readily venting gases formed during chemical cleaning.

3.12.1 Easy access shall be provided to top and bottom, tube sheets and demister pads. Where overhead clearance is minimal (that is, clearance does not allow for overhead rigging) the head covers shall be in sections. These sections shall be such that two men, without aid of ❑echanical advantage, can remove the

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covers . Flanges, covers, plates and partitions that must be removed for performing maintenance shall be easily accessible with no burning, cutting, or welding of these components permitted in gaining access.

3.13 Thermal insulation. Unless otherwise specified, the unit shall be insulated in accordance with MIL-STD-769 (see 6.2).

3,14 Tinning. When specified (see 6.2), all metal in contact with the distillate shall be tinned with pure tin (see 6.3 and appendix B).

3.15 Electrical equipment. “

3,15,1 Electrical equipment shall be in accordance with MIL-E-917.

3,15.1.1 Unless otherwise specified (see 6.2), electrical equipment shall be designed for operation from type I, 440 volt power as defined in accordance with MIL-STD-1399, section 300.

3.15,1.2 Switches. safety controls. Pressure switches, manually operated switches , controllers , and master switches, and relays furnished as components shall be wired with cable conforming to MIL-C-915. These components, except the pressure switch, may be mounted in a single control cabinet, class 1, drip-proof in accordance with MIL-E-2036. A set of safety controls shall be provided to shutdown a malfunctioning unit to prevent damage to the unit. The safety control system shall be of the type that will indicate the control signal causing the shutdown.

3.15.2 Installation. Care shall be taken to ensure that installation of electrical apparatus is mounted away from possible sources of water, oil, or moisture damage. Clearance shall be provided in the immediate vicinity of electrical equipment to permit and facilitate maintenance.

3.16 Fasteners. Fasteners shall be nickel-copper alloys in accordance with MIL-S-1222 except foundation bolting which may be zinc plated carbon steel in accordance with MIL-S-1222.

3.16.1 Screw threads. Thread form dimensions and tolerances shall be in accordance with FED-STD-H28.

3.16,2 Threaded parts. Cap screws shall not be used. Tapered pipe threads are unacceptable except for use in grease fittings located on nonmoving parts. Threaded fasteners shall be in accordance with table I. A class 5 interference fit conforming to ANSI B1.12 may be substituted for assembly of tap-end of studs The practice of “bottoming” or “shouldering” studs shall not be used. For the set end of studs, a class 3A fit used with sealing compound, in accordance with grade AV or AW of MIL-S-22473, may be substituted for a class 5 interference fit. Threaded fasteners with torque requirements shall be lubricated before assembly.

(a) For class A units, lubricant shall be graphite in isopropanal, per MIL-L-24131 (Military Symbol CGI).
(b) For class B units, the lubricant per MIL-A-907.

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3.16.2.1 Collar or stud bolts. Where collar bolts or stud bolts are used to make up a multiple, flange joint, so that it can be selectively disassembled, such bolts or studs shall have a square extension beyond *the* threads on one end, for use of a wrench to prevent turning of the bolt when the nuts are tightened or removed.

3.16.2.2 Thread engagement. Stud engagement shall be sufficient to hold the maximum design load. In no case shall engagement be less than the nominal diameter of the fastener. Nuts shall be fully engaged in all cases. For a threaded fastener, not less than one thread but not more than four threads shall protrude beyond the crown of the nut.

3.17 Piping, valves and fittings. Piping, valves, fittings, and associated piping components shall be in accordance with MIL-STD-777 for surface ships, and in accordance with MIL-STD-438 for submarines except as modified herein (see 6.2). In addition to brazing flanges allowed in accordance with MIL-STD-777, welded copper-nickel flanges of either the slip-on or weld-neck design are acceptable provided that interface geometry is not affected. Root connections shall be in accordance with Drawing B-214 to minimize stress connections and provide increased wall thickness. Methods and types shown on Drawing B-214 illustrate the desired principles. The location and general arrangement of piping shall provide access for take down and maintenance.

3.17.1 Flanges. Flanges for pipe connections shall be 150 lb/in^2 in accordance with MIL-F-20042, except flanges in submarine installations shall be as specified (see 6,2).

3.18 Welding and allied processes. Fabrication and inspection of welded and silver brazed piping joints shall be in accordance with MIL-STD-278.

3.18.1 Casting inspection and repair. Casting inspection and repair shall be in accordance with MIL-STD-278.

3.19 General features.

3.19.1 Primary requirements. Characteristics not normally incorporated in the design of commercial equipment shall be required on Naval ships. Primary requirements which are the basis for the design of Naval equipment shall be as follows:

(a) Mean-time-between-failure (MTBF) not less than 2,000 hours

When: $\text{MTBF} - \frac{\text{Total operating hours}}{\text{No. of failures that prevent the unit from performing its mission.}}$

(b) Mean-time-to-repair (MTTR) not more than 12 hours.
(c) Time between overhauls not less than 5 years.

3.19.2 Standard stock items. Standard stock materials, parts, and hardware shall be used. For the purpose of this specification, standard stock is defined as material listed in the Federal Supply Catalog and includes stock items such as bearings, grease, oil, wire, bolts, screws, nuts, and washers. (C o p i e s

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of this catalog may be consulted in the office of the Defense Contract Administration Services Management Area (DCASMA)). Proprietary parts or parts available from only one source shall not be used where equivalent or similar parts are available as standard parts or are available from two or more sources.

3.19.3 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.20 Identification plates. Each unit shall bear an identification plate in accordance with type A, B, C, or D of MIL-P-15024 and MIL-P-15024/5 with material choice being limited to the wrought brass, cast brass, cast bronze, or corrosion-resisting steel prescribed therein. The plates shall be normal service plates. Provision shall be made for the following information:

(a) Name of manufacture.
(b) Contract or order number. (Utilize width of plate to allow maximum number of spaces.)
(c) National stock number (allow 20 spaces),
(d) Date of manufacture.
(e) Blank space for DCASMA stamp.
(f) Blank space for unit number (2 spaces, stamped by shipyard).

3.21 Painting. painting of external nonferrous castings, piping or other parts is not required. External unmachined surfaces of ungalvanized ferrous parts shall be thoroughly cleaned and coated with two coats of heat resisting paint in accordance with TT-P-28 or its commercial equivalent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program (see 6.3). The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

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(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examinations and tests as specified in table III (see 6.3) .

TABLE III. First article inspection.

Examination or test	Requirement	First "article test
Power requirements	3.5 "	4.6.1
Inclination	3.3.9	4.6.2
Salinity content	384.3	4.6.3
Capacity	3.4.1, 3.4.2, 3.4,4, 3.4.5	4.6.4, 4.6.4.1
Heating units: Dielectric voltage Insulation resistance	353.1 3.3.1.2 3.3.1.3	 4.6.5 4.6.5
Hydrostatic pressure	3,3.7	4.6.6
Shock	3,6.1	6.6.7
Vibration	3.6.2.2	4.6.8
Noise	3.6.2.1	4.6.9
Maintainability	3.19.1	4.6.10
Automatic operation	3.3.8	4.6.11
Reliability	3.19.1	4.6.12

4.4 Quality conformance inspection. Quality conformance inspection shall be performed on all distilling units and shall consist of applicable examinations and tests specified in table IV, except where modified by the contract or order (see 6.2 and 6.3).

TABLE IV. Quality conformance inspection.

Examination or test	Requirement	Quality conformance test
Salinity content	3.4.3	4.6.3
Performance	3.4.4, 3.4.1	4.6.4, 4.6.4.1
Heating units	3.3.1	4.6.5

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4.5 Examination.

4.5.1 General. The completed unit shall be thoroughly examined to determine conformance to this specification. Drawings shall be checked with respect to material, finish, construction, assembly, dimensions, weight and marking of identification and description plates. This examination shall be limited to that which can be performed without disassembling the unit in such a manner that its performance, durability, or appearance would be affected.

4.6 Tests.

4.6.1 Power test. The unit shall be tested to determine maximum power required to operate the distiller. Maximum power is defined as all electric components and all immersion heaters operating.

4.6.2 Inclination test. Unless otherwise specified (see 6.2), the unit shall be operated to demonstrate ability to meet capacity requirements as specified in 3.4.1, and when subject to roll and pitch conditions specified in 3.3.9, rocking platform requirements shall be as specified.

4.6.3 Salinity content. The salinity content of the distillate shall meet the requirements specified in 3,4.3. The salinity content shall be tested using the salinity indicating equipment that is installed on the unit. If accuracy is questionable by the methods available aboard the ship, a sample of the distillate shall be submitted to a qualified laboratory for determination of the salinity content in accordance with ASTM D 512.

4.6.4 Capacity. Each distilling unit shall be tested in the manufacturer's plant, after installation prior to preliminary trials and during official preliminary trials, to demonstrate clean tube capacity as specified in 3.4.5 and that all component parts are in operating condition. The test conducted after installation and prior to preliminary trials shall use feed water available at the ship's berth. Should it be considered that the water available is such as to foul the tubes of the unit, this test may be waived or postponed by the Government. The seawater used for feed for the preliminary trials test shall be not less than 35,000 p\m total dissolved solids. For new designs and vendor first article testing, a 90-day test on natural seawater is required.

4.6.4.1 An additional capacity test shall be made before final acceptance of the ship. Starting with clean tubes, the unit shall be operated for of 250 hours , with feedwater of 35,000 p/m. The density of brine overboard shall not exceed the maximum specified by the manufacturer in the technical manual for the unit , It shall be demonstrated that the average capacity over the specified period of operation is not less than the specified rated capacity (see 3.4.1). The salinity content of the distillate shall not exceed that specified in 3.4.3 The compressor motor shall not be overloaded during the specified period of operation.

4.6,5 Heating units. Heating units shall conform to the requirements of 3.3.1 and shall be tested in accordance with MIL-H-22577.

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4.6.6 Hydrostatic pressure. All parts of the distilling unit shall be subjected to a hydrostatic pressure test as follows:

Name of part	Test Pressure , lb/in^2 gauge
Evaporator and vapor compressor	10
Heat exchangers and other parts not specified	1-1/2 times working pressure (50 minimum)

4.6.7 Shock, The assembled unit shall meet the shock requirements specified in 3.6.1 and shall be shock tested in accordance with MIL-S-901.

4.6.8 Vibration. The unit and all components shall meet the vibration requirements specified in 3,6.2.2 and shall be tested in accordance with MIL-STD-167-1.

4.6.9 Noise. The unit shall meet the noise requirements specified in 3.6.2.1 and shall be tested in accordance with MIL-STD-740-1 and MIL-STD-740-2.

4.6.10 Maintainability. The unit shall meet the requirements for maintainability as specified in 3.19.1.

4.6.11 Automation operation. The unit shall meet the automatic operation specified in 3.3.8 within the time requirement specified in 3.4.4.

4.6.12 Reliability. The unit shall meet the primary requirements as specified in 3.19,1. When specified (see 6.2), a reliability program shall be implemented in accordance with the requirement tasks of MIL-STD-785.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.7.)

5.1 Preservation. Preservation shall be level A or commercial as specified in the contract or order (see 6.2).

5.1.1 Level A. The distillation unit shall be preserved in accordance with method III of MIL-P-116. Interior surfaces shall be thoroughly dried after which openings shall be sealed with caps, plugs, or plywood blanks to prevent entry of foreign matter. Barrier material in accordance with grade A of MIL-B-121 shall be placed between the plywood blank and the mating flange.

5.1.1.1 Accessories . Gauges and thermometers shall be removed from the unit and individually preserved by methods IA and III, respectively, in accordance with MIL-P-116,

5.1.2 Commercial. The distillation unit shall be preserved in accordance with ASTM A 395.

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5.2 Packing. Packing shall be level A, B or commercial as specified in the contract or order (see 6.2) .

5.2.1 Levels A. B and C. The distillation unit shall be packed in a sheathed crate in accordance with MIL-C-104, with type, class and style selection at the option of the contractor. When specified (see 6.2), for levels B and C only, packing shall be in unsheathed crates in accordance with MIL-C-132 or - - MIL-C-3774. The closure and strapping of containers and the anchoring, blocking, bracing, cushioning, and waterproofing of container contents shall be in accordance with the applicable container specification or appendix thereto.

5.2.1.1 Detached accessories. Detached accessories packaged as specified in 5.1.1.1 shall be packed in boxes in accordance with class 1 of PPP-B-621 or PPP-B-601 (domestic) and secured within the shipping containers with the complete unit .

5.2.2 Commercial. Distillation units, preserved as specified in 5.1 shall be individually packaged in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required (see 6.2), for levels A, B, C (see 6.2), interior and exterior shipping containers shall be marked in accordance with MIL-STD-129, commercial containers in accordance with ASTM D 3951.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Distillation units of the thermocompression type will be a permanent installation on Naval ships and used to produce fresh water from seawater.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Whether a particular type is required (see 1.2.1).
(c) Class required (see 1.2.2).
(d) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(e) When first article inspection is required (see 3.1).
(f) Whether the units shall be furnished “knock down” or “package” type (see 3,3).
(g) Whether resilient mounts are to be provided with unit (see 3.3).
(h) If feed pump is required (see 3.3(g)) and feedwater pressure.
(i) Voltage and current required for electric heaters (see 3.3.1).
(j) Whether thermometers are to be furnished with the distillation units (see 3.3.6 and 6.5.2(f)).
(k) Ship motion and attitude performance requirements (see 3.3.7) .
(1) Rated capacity required (see 3,4.1).

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(m) Continuous or intermittent injection of feedwater treatment chemicals (see 3.4.1 and 3.4.2).
(n) Distillate temperature if other then specified (see 3.4.l(d)).
(0) Quantity of units per ship and number of ships.
(p) Purity of distillate - whether nuclear or non-nuclear (see 3.4.3).
(q) Shock requirements if other than specified (see 3.6.1).
(r) Noise and vibration requirements if other than specified (see 3.6.2).
(s) Whether ball bearings shall be in accordance with MIL-B-17931 (see 3.6.2.3).
(t) Type of feed heat exchanger, shell and tube or plate and frame (3.8).
(u) Structureborne noise levels for meters, if other than specified (see 3.10.3).
(v) Whether thermal insulation is to be provided with the unit (see 3.13).
(w) Whether surfaces in contact with distillate (required only on diesel driven submarines) are to be tinned (see 3.14).
(x) Power supply for electrical equipment, if other than specified (see 3.15.1.1),
(y) Submarine piping requirements (see 3.17 and 3.17.1).
(z) Inclination testing requirements for larger units (see 4.6.2).
(aa) Whether a reliability program is required (see 4,6.12).
(bb) Whether quality conformance inspection requires modification (see 4.7).
(cc) Selection of applicable levels of preservation, packaging, and packing (see 5.1, 5.2, and 5.2.1).
(old) Special marking required (see 5.3).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements , a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.1.1 and appendix A	DI-MISC-80678	Certificate data/report	---
3.14	DI-E-20477	Imaged aperture\ tabulating cards	---
3.14 and appendix B	DI-DRPR-80651	Engineering, drab'ings	---
3.19.1	DI-R-24039	Report, design	
4.1.1	DI-R-4803	Inspection system program plan	---
4.3	DI-T-4902	First article inspection report	---
4.4	UDI-T-23732	Procedures, test	---
	DI-T-5329	Inspection and test report	---

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The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Technical manuals. The requirements for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which in included as an exhibit to the contract. The technical manuals must be acquired under separated contract line item in the contract.

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.6.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.6.2 On board repair parts, On board repair parts are those parts which should be carried on board Naval ships for routine maintenance of parent equipment by shipboard personnel. On board repair parts should be furnished with the distillation unit specified herein and should include the following:

(a) 4 set of V-belts.
(b) 100 percent gaskets.
(c) 100 percent gauge, sight, and flowmeter glasses (if used) .
(d) 1 electric heater.
(e) 1 gauge of each size and type.
(f) 1 thermometer of each size and type (to be furnished by manufacturer unless otherwise specified (see 6.2).

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(g) 10 percent of special tubes for heat exchanger (if included in the design), including tube packing.
(h) 1 ammeter of each size and type.
(i) Repair parts for motors, controllers, switches, and relief valves should be furnished in accordance with the requirements of the applicable specifications for such components.
(j) 10 percent of compression type heat exchanger tube fittings (if used) .
(k) Repair parts for feedwater pressure and flow regulating valve.
(1) Phenolic tube plugs to plug 10 percent of evaporator tubes.

6.7 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.8 Drawing review activity. As used herein "drawing review activity" is generally a Government command or agency such as NAVSEA, or an authorized representative . Communication with the drawing review activity should be handled through the contracting activity.

6.9 Lubricants and compounds. Any exception to 3.2.6 must be approved by NAVSEA.

6.10 NAVSEA approval . NAVSEA approval is required for any exceptions to 3.4.2.1.1, 3.4.2.1.2 and 3.4.2.2.

6.11 Subject term (key word) listing.

Electric immersion heater
Flow control
Flow meter
Rotameter
Totalizing meter

6,12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4620-N035)

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APPENDIX A

CERTIFICATION DATA/REPORT TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical content requirements that should be included on certification data/reports when required by the contract or order. This appendix is mandatory only when data item description DI-MISC-80678 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30, CERTIFICATION DATA/REPORT

30.1 Certification data/report. When required by the contract or order, certification shall be required for production units where first article testing is not performed. The certificate shall contain sufficient information to assure the Government that the contractor has met or exceed all the requirements of the contract and applicable industry association/society standards. This data shall include, but not be limited to, specifying performance characteristics and the suitability of the equipment to meet its intended use.

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APPENDIX B

ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical content requirements that should be included on drawings when required by the contract or order. This appendix is mandatory only when data item description DI-DRPR-80651 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. DRAWINGS

30.1 Drawings. When required by the contract or order, drawings shall contain the following information.

30.1.1 Outline drawing. An external arrangement drawing shall show all necessary external views of the unit and shall include all external dimensions required for reproduction on ship's machinery arrangement drawings. These drawings shall be guidance for the shipyard in designing the foundation structure for the unit, for installation thereof. It shall show connection of the unit to the external piping, The drawing shall show the space required for removal and replacement of tubes in the heat exchanger components, removal of waterboxes , covers of access openings, zinc anodes, and vapor separators. Lifting lugs or eyebolts, with their dimensional openings, provided for lifting the unit or its components shall be shown on the drawing.

30.1.2 Assembly drawing. A drawing showing complete longitudinal and transverse cross sectional views of the unit which shall show the relationship of all parts, arrangement of tubes in heat exchanger components, method of tube end expansion, baffles, vapor separators, and welding of adjacent parts. Liberal use of enlarged views or sections shall be made. Sub-assembly drawings conforming to the above shall be furnished to aid in the clarity of individual components . The drawings shall be such that a thorough understanding of the design and construction of the apparatus may be obtained without reference to related detailed drawings.

30.1.3 List of materials. The assembly drawing shall contain a list of materials showing names of parts with identifying numbers and materials of all parts . The identifying numbers shall also be shown adjacent to the part depicted in the various views, with arrows pointing to the parts. Lubricants required for threaded parts shall be identified (see 3.16.2).

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30.1.3.1 Commercial material option. If the manufacturer substitutes commercial material and such substitution has been approved by the design review activity, he shall include two columns in the list of materials on his drawings (see 3.2). One column shall be used for the specified Federal or Military specifications , and the other shall show the commercial specification number for the substituted material. Notes shall then appear on the drawings stating that the materials listed in the commercial specification column have physical and chemical properties which are equal to or better than the specified Federal or Military specification and are being substituted by the manufacturer.

30.1,4 Detail Drawings. Detail drawings of all major parts such as shells, waterboxes , tube sheets, tube supports, baffles, vapor separators, and demisters. The drawings shall be completely dimensioned, with finishes and welding symbols indicated, as required for manufacture,

30.1.5 Piping arrangement drawing. A diagrammatic piping arrangement drawing shall show complete piping required for the operation of the unit. Piping furnished with the unit and piping furnished by the shipbuilder shall be clearly indicated. Location of gauges, thermometers, valves, orifice plates, acid cleaning connections, and salinity indicators shall be indicated.

30.1.6 Wiring drawing. Wiring assembly drawings showing terminal connections shall be provided for electrical components furnished with this unit.

30.1.7 Certification data drawings. Certification data sheets shall be prepared and shall include a tabulation of the following data:

(a) Velocity of circulating water under rated conditions in the tube of each component.
(b) Complete heat balance diagrams showing capability of the unit to produce rated capacity over the required operating range; diagrams for seawater temperatures of 28, 60, and 85°F are required.
(c) Hydrostatic test pressure for *all* components, list of on board repair parts furnished and all similar pertinent design data
(d) A drawing list tabulation with columns for drawing title, manufacturer's drawing number, and revision symbol. This list shall include all equipment drawings which comprise the design. The revision symbol column shall be kept up to date to the time of manufacture so that it will indicate for each drawing the revision applicable to the equipment as built.

30.1.8 Timing. Drawings shall include tall information as to method of timing used and the resulting thickness (see 3.14).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-16196E(SH)	DISTILLATION UNIT, WATER, THERMOCOMPRESSION

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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